

Q1 JEE Main 2020 - 7 Jan (Morning)

Total number of 6-digit numbers in which only and all the five digits 1, 3, 5, 7 and 9 appear, is

(A) 5^6

(B) $\frac{1}{2}(6!)$

(C) $6!$

(D) $\frac{5}{2}(6!)$

Q2 JEE Main 2020 - 8 Jan (Morning)

An urn contains 5 red marbles, 4 black marbles and 3 white marbles. Then the number of ways in which 4 marbles can be drawn so that at the most three of them are red is

Q3 JEE Main 2020 - 8 Jan (Evening)

The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word **EXAMINATION** is

Q4 JEE Main 2020 - 9 Jan (Morning)

If number of 5 digit numbers which can be formed without repeating any digit while tenth place of all

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of the numbers must be 2 is $336 \cdot k$ find value of k

(A) 8

(B) 7

(C) 6

(D) 5

Q5 JEE Main 2020 - 2 Sep (Morning)

If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'MOTHER' is

Q6 JEE Main 2020 - 2 Sep (Evening)

Let $n > 2$ be an integer. Suppose that there are n Metro stations in a city located along a circular path. Each pair of stations is connected by a straight track only. Further, each pair of nearest stations is connected by blue line, whereas all remaining pairs of stations are connected by red line.

If the number of red lines is 99 times the number of blue lines, then the value of n is :

- (A) 199
 (B) 201
 (C) 101
 (D) 200

Q7 JEE Main 2020 - 3 Sep (Evening)

The total number of 3 -digit numbers, whose sum of digits is 10, is

Q8 JEE Main 2020 - 4 Sep (Evening)

A test consists of 6 multiple choice questions, each having 4 alternative answers of which only one is correct. The number of ways, in which a candidate answers all six questions such that exactly four of the answers are correct, is

Q9 JEE Main 2020 - 5 Sep (Morning)

The number of words, with or without meaning, that can be formed by taking 4 letters at a time from the letters of the word 'SYLLABUS' such that two letters are distinct and two letters are alike, is

Q10 JEE Main 2020 - 5 Sep (Evening)

There are 3 sections in a question paper and each section contains 5 questions. A candidate has to answer a total of 5 questions, choosing at least one question from each section. Then the number of ways, in which the candidate can choose the questions, is :

- (A) 2250
 (B) 3000
 (C) 1500
 (D) 2255

Q11 JEE Main 2020 - 6 Sep (Morning)

Two families with three members each and one family with four members are to be seated in a row. In how many ways can they be seated so that the same family members are not separated?

- (A) $2! 3! 4!$
 (B) $(3!)^3 \cdot (4!)$
 (C) $(3!)^2 \cdot (4!)$
 (D) $3!(4!)^3$

Q12 JEE Main 2020 - 6 Sep (Evening)

The number of words (with or without meaning) that can be formed from all the letters of the word "LETTER" in which vowels never come together is

Answer Key

Q1 (D)	Q2 (490)	Q3 (2454)	Q4 (A)
Q5 (309)	Q6 (B)	Q7 (54)	Q8 (135)
Q9 (240)	Q10 (A)	Q11 (B)	Q12 (120)

Q1

1, 3, 5, 7, 9

For digit to repeat we have 5C_1 choice

And six digits can be arrange in $\frac{6!}{2!}$ ways.

Hence total such numbers = $\frac{5 \cdot 6!}{2!}$

Q2

0 Red, 1 Red, 2 Red, 3 Red

Number of ways = ${}^7C_4 + {}^5C_1 \cdot {}^7C_3 + {}^5C_2 \cdot {}^7C_2 + {}^5C_3 \cdot {}^7C_1$
 $= 35 + 175 + 210 + 70 = 490$

Q3

EXAMINATION

2N, 2A, 2I, E, X, M, T, O

Case I All are different so ${}^8P_4 = \frac{8!}{4!} = 8 \cdot 7 \cdot 6 \cdot 5 = 1680$

Case II 2 same and 2 different so ${}^3C_1 \cdot {}^7C_2 \cdot \frac{4!}{2!}$
 $= 3 \cdot 21 \cdot 12 = 756$

Case III 2 same and 2 same so ${}^3C_2 \cdot \frac{4!}{2! \cdot 2!} = 3 \cdot 6 = 18$

$\therefore$ Total = $1680 + 756 + 18 = 2454$

Q4

Number of numbers = $8 \times 8 \times 7 \times 6 = 2688 = 336k$

$\Rightarrow k = 8$

Q5

EHMORT in alphabetical order

E 5!
 H 4!
 M E 4!
 M H 4!
 M O E 3!
 M O H 3!
 M O R 3!
 M O T E 2!
 M O T H E R 1

$$\text{Rank} = 2 \times 5! + 2 \times 4! + 3 \times 3! + 2! + 1$$

$$= 309$$

Q6

Number of two consecutive stations = n

Number of two non-consecutive stations = ${}^nC_2 - n$

Now, According to the question,

$$\Rightarrow {}^nC_2 - n = 99n$$

$$\Rightarrow \frac{n(n-1)}{2} - 100n = 0$$

$$\Rightarrow n - 1 - 200 = 0$$

$$\Rightarrow n = 201$$

Q7

$$x + y + z = 10, x \geq 1, y \geq 0, z \geq 0$$

$$\text{Let } x - 1 = x'$$

$$x' + y + z = 9, x', y, z \geq 0$$

$$\text{Number of solutions are } {}^{9+3-1}C_2 = {}^{11}C_2$$

$$= 55$$

But for $x' = 9 \Rightarrow x = 10$ which is not possible.

$\therefore$ Total required numbers = $55 - 1$

$$= 54$$

Q8

Select any 4 questions in 6C_4 ways which are correct

Number of ways of answering wrong question = 3

: Required number of ways = ${}^6C_4 \times 3^2 = 135$

Q9

LLSSYABU

For two alike and two distinct letters, select any one pair from LL, SS in 2C_1 ways

Now from rest, select any 2 in 5C_2 ways and

they can be arranged in $\frac{4!}{2!}$ ways

$\therefore$ Required number of ways = ${}^2C_1 \times {}^5C_2 \times \frac{4!}{2!}$
= 240

Q10

Each section has 5 questions.

$\therefore$ Total number of selection of 5 questions

$$= 3 \times {}^5C_1 \times {}^5C_1 \times {}^5C_3 + 3 \times {}^5C_1 \times {}^5C_2 \times {}^5C_2$$

$$= 3 \times 5 \times 5 \times 10 + 3 \times 5 \times 10 \times 10$$

$$= 750 + 1500$$

$$= 2250$$

Q11

$$\text{No. of arrangement} = ({}^3P_3 \times {}^3P_3 \times {}^4P_4) \times {}^3P_3$$

$$= ({}^3P_3)^3 \times {}^4P_4$$

Q12

For vowels not together

1st arrange L, T, T, R in $\frac{4!}{2!}$ ways

Then put both E in 5 gaps formed in 5C_2 ways

$$\therefore \text{No. of ways} = \frac{4!}{2!} \cdot {}^5C_2 = 120$$